

Name: _____

Class: _____

ACTIVITY SHEET

Chapter 1: Prior knowledge

Complete the following statements by filling in the missing words, all of which can be found on the word list below. (Hint: some words are used more than once.)

Word list

atoms	groups	nucleus	shells
chemical	heavier	occupied	six
eight	isotopes	periodic	speed
electrons	least	periods	symbol
element	lighter	positive	two
equal	mass	protons	uncharged
fixed	negative	radioactivity	
full	neutrons	share	

- All matter is composed of extremely small particles called _____.
- Atoms have an internal structure consisting of a tiny, dense central area called the _____, surrounded largely by empty space.
- Atoms are formed out of three types of sub-atomic particles: _____, _____ and _____.
- Protons and neutrons are _____ sub-atomic particles found inside the nucleus.
- Electrons are _____ sub-atomic particles found outside of the nucleus.
- Protons and neutrons, which are densely packed together in the nucleus, provide most of an atom's _____.
- Electrons, which move around the nucleus at high _____, occupy most of the volume of the atom but provide little of its mass.
- A proton has one _____ charge and an electron has one _____ charge, but a neutron is _____.
- There are 93 different types of naturally occurring atoms. Each type is defined by the number of _____ in its nucleus. For example, every carbon atom has six of these sub-atomic particles.
- Uncharged atoms have _____ numbers of protons and electrons, so an uncharged carbon atom will have _____ electrons.

- 11 A substance consisting of only one type of atom is called an _____. For example, the element carbon consists of atoms with six _____ only.
- 12 Each element is given a unique name and _____; e.g. magnesium, symbol Mg.
- 13 Each type of atom can have varying numbers of _____ in the nucleus. For example, carbon atoms can have 6, 7 or even 8 of them. These different forms of an atom are known as _____.
- 14 Some isotopes are unstable and emit _____ when their nuclei break down to form more stable isotopes.
- 15 An atom's electrons have discrete amounts of energy and are located in specific energy _____ around the nucleus, depending on how much energy they have. Those in the innermost shells have the _____ amount of energy.
- 16 Each shell can accommodate up to a _____ number of electrons. The first shell accommodates up to _____, the second and third shells up to _____ each.
- 17 Atoms are stable if their outermost occupied shell is _____. Atoms may _____ or transfer electrons in order to reach stable energy states.
- 18 The _____ table is an arrangement of the elements in order of the number of protons in the nuclei of their atoms going from left to right across the rows.
- 19 The rows are called _____ and the columns are called _____.
- 20 Those elements in the same group have similar _____ properties, because how atoms react depends on how many electrons are in the outermost _____ shell.

Having difficulty?

Visit the BBC Bitesize website on 'Atoms' and explore pages 1, 2 and 4.